

KITEWRIGHT LIFT · REQUIREMENTS

Kitewright Lift requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-lift		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Status of each requirement against KWL-CAL-001 for the constructable design; R11 stated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Lift requirements

Requirements for the first prototype, with the status of each against the TRL 3 calculations (KWL-CAL-001) for the constructable design (KWL-DDR-002). Targets are as set at TRL 1; none has been changed. "Met on paper" means the calculation shows the target is reachable; every requirement is verified by test at TRL 4 or later. Requirements not met or at risk are posed to Amish in docs/REVIEW.md and listed in the design decisions register (docs/06-design-decisions.md).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (KWL-CAL-001)
R1	Maximum take-off mass	25 kg or less including payload	Weigh the fully loaded aircraft	Not met with a 5 kg payload: 25.5 kg. Ready to fly 20.5 kg. Open decision 1
R2	Payload at sea level	5 kg on the core mount (target)	Hover test with calibrated mass	At risk: thrust is ample (thrust to weight 2.83) but R1 leaves 4.5 kg. Open decision 1
R3	Payload at altitude	2 kg at air density equivalent to 5,000 m (target)	AltiRig thrust data, then flight at a high site	Met on paper: thrust to weight 1.94 at 5,000 m and -20 °C with 2 kg
R4	Hover time	At least 20 min with 5 kg at sea level and 10 min with 2 kg at 5,000 m and -20 °C (targets)	Timed hover flights with logged energy use	Not met: 8.9 min and 7.1 min with the LiFePO4 ColdCell packs. Open decision 2
R5	Motor-out tolerance	Controlled descent and landing after one motor failure at rated payload	Deliberate motor cut test over a safe area	Met on paper: thrust to weight with one motor out 2.20 at sea level, 1.51 at 5,000 m; yaw control to be shown in the motor-cut test
R6	Wind	Holds position within 2 m in 10 m/s steady wind (target)	Logged position hold on a windy test day	Met on paper for thrust: 18 N drag, 4.2 deg tilt; position hold depends on tuning and is shown in flight
R7	Tethered endurance	At least 2 h continuous hover on fixed-voltage tether power (target)	Tethered endurance run with logged voltages and temperatures	Met on paper: 3.0 kW hover on a 4 kW converter (margin 1.35); converter and motor temperatures over 2 h to be measured
R8	Float and line drop	Places a float within 3 m of a target from 10 m height (target)	Drop trials over water	Met on paper: expected miss 2.2 m in 10 m/s wind when the drift is aimed off
R9	Transport	Folds to fit a case of 1.2 m by 0.6 m by 0.5 m or smaller; set up in 10 min or less (target)	Pack and timed set-up trials	Met on paper: folded 0.84 x 0.48 x 0.45 m with the gear on; set-up about 9 min
R10	Operating temperature	Flies from -20 to +45 °C with ColdCell packs (target)	Cold soak then hover test	Met on paper by part ratings; pre-heat about 53 Wh from the ground supply
R11	Prototype cost	USD 5,000 or less excluding payloads	Bill of materials and receipts	Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 5,739 (USD 739 over the target)

■ Assumptions

- Air at 5,000 m and -20 °C: 54.0 kPa and 0.743 kg/m³ (71 % of the pressure and 61 % of the density at sea level).
- Motor and propeller performance from makers' figures for the 30 inch, 100 KV class, until AltiRig measures it at 5,000 m density.

- ColdCell delivers 85 % of its rated energy at -20 °C with its heater running (ColdCell R1).
- The Kitewright Core stack with its payload mount weighs 1.0 kg or less (Core R9).
- Local rules at the test sites allow external loads and tethered flight with permission.

■ Safety

Safety: The requirements describe a 25 kg class aircraft with large propellers, lithium iron phosphate packs, a 400 V tether and suspended loads. R5 (motor out) and R7 (tether) are tested only over cleared ground with the keep-out and isolation set out in the build plan's safety stops (KWL-BLD-001, section 6).